实验三　参考程序

1、

```
I=imread('image.tif');
J = imnoise(I,'gaussian',0.02);          %添加高斯噪声
ave1=fspecial('average',3);              %产生 3×3 的均值模版
ave2=fspecial('average',5);              %产生 5×5 的均值模版
K = imfilter(J, ave1,);                  %均值滤波 3×3
L = imfilter(J, ave2);                   %均值滤波 5×5
imshow(I);
figure,imshow(J);
figure,imshow(K);
figure,imshow(L);
```

2、利用 Roberts 锐化对 lena 图像进行增强处理

```
I=imread('lena.tiff');
grayPic=rgb2gray(I);
figure(1);
imshow(grayPic);
[high,width]=size(grayPic);
newGrayPic=grayPic;
robertsNum=0;
robertThreshold=0.8;
for j=1:high-1
for k=1:width-1
robertsNum = abs(grayPic(j,k)-grayPic(j+1,k+1)) + abs(grayPic(j+1,k)-grayPic(j,k+1));
newGrayPic(j,k)=robertsNum;
end
end
figure(2);
imshow(newGrayPic);
```

3、利用 Sobel 锐化对图像进行增强处理

```
I=imread('lena.tiff');
I=rgb2gray(I);
[L,R]=size(I);
Y=double(I);
for i=2:L-1
    for j=2:R-1
        dx=(Y(i+1,j-1)-Y(i-1,j-1))+2*(Y(i+1,j)-Y(i-1,j))+(Y(i+1,j+1)-Y(i-1,j+1));
        dy=(Y(i-1,j+1)-Y(i-1,j-1))+2*(Y(i,j+1)-Y(i,j-1))+(Y(i+1,j+1)-Y(i+1,j-1));
        g(i,j)=sqrt(dx^2+dy^2);
    end
end
```

```
figure(1)
imshow(I);
figure(2)
imshow(uint8(g));
```

4、利用拉普拉斯锐化对图像进行增强

```
clear
clc
I=imread('lena.tiff');
subplot(1,2,1)
imshow(I)
I=im2double(I);
[m,n,c]=size(I);
A=zeros(m,n,c);
```

```
subplot(1,2,2)
 imshow('lena.tif')
```